

Predicting Severity of Mental Health Symptoms Using Behavioral Data And Brain Scans

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Topic of Interest

Quantitative brain imaging features and tabular behavioral/clinical assessment data are typically analyzed separately when studying mental health.

We aim to explore whether mental health symptom severity can be best predicted best using brain imaging features, behavioral/clinical data, or a combination of both.

We focus on symptom severity rather than diagnosis because people with the same diagnosis can still experience very different levels of symptoms.

Summary: 1. Why Is This Topic Important?

1.

Mental health conditions are common, complex, and difficult to capture with diagnosis alone, so predicting symptom severity may provide a more detailed understanding of patient needs and help improve monitoring, treatment planning, and research on brain–behavior relationships.

Summary: 2. How Will Data Be Used? 3. Potential Impact

2.

- Transdiagnostic Connectome Project dataset with 241 participants
- Includes resting-state fMRI-derived connectivity features data, demographics, and 50+ behavioral/clinical assessments
- Models:

Behavioral Data —▶ Random Forest —▶ Prediction
Imaging Features —▶ Random Forest —▶ Prediction
Behavioral + Imaging —▶ Random Forest —▶ Prediction

3.

- Findings could show whether combining brain scans with behavioral data improves symptom prediction, and whether brain imaging adds useful information beyond questionnaires and clinical measures alone

Compare MAE • RMSE • R^2

Machine Learning Algorithm(s)

- Random Forest Regression as the main machine learning
- Random Forest is a strong fit because it can handle nonlinear relationships, works well with tabular and imaging-derived features, and provides feature importance measures that can help show which variables contribute most to symptom severity prediction
- Use the model to compare:
 - Behavioral-only Random Forest
 - Imaging-only Random Forest
 - Multimodal Random Forest

Research Question

- Can mental health symptom severity be better predicted from brain imaging features, behavioral/clinical data, or a combination?
- Does a multimodal model outperform models that use only imaging data or only behavioral data?

Dataset

- **Name:** The Transdiagnostic Connectome Project
- **Source:** OpenNeuro
- **Link:** <https://openneuro.org/datasets/ds005237/versions/1.1.3>
- **Formal Citation** (via OpenNeuro website):
 - Sidhant Chopra, Carrisa V. Cocuzza, Connor Lawhead, Jocelyn A. Ricard, Loïc Labache, Lauren Patrick, Poornima Kumar, Arielle Rubenstein, Julia Moses, Lia Chen, Crystal Blankenbaker, Bryce Gillis, Laura T. Germine, Ilan Harpaz-Rote, BT Thomas Yeo, Justin T. Baker, and Avram J. Holmes (2025). Transdiagnostic Connectome Project. OpenNeuro. [Dataset] doi: doi:10.18112/openneuro.ds005237.v1.1.3

Training & Testing Considerations

- The dataset contains 241 participants with multiple types of data per person, including MRI/fMRI data, demographics, and more than 50 behavioral and clinical assessments.
- Because this is a moderate sample size, using Random Forest Regression on extracted imaging features and tabular clinical data rather than training a deep learning model on raw scans is more appropriate
- Use 5-fold cross validation to improve reliability and model stability
 - 4 folds for training, 1 for validation
- Patients would be split:
 - 80% training = 192 participants
 - Split into 5 folds (~38-39 participants each)
 - 10% testing = 49 participants

Sources of Bias

#1: Group Underrepresentation

The dataset contains an unequal distribution of sex, with 127 Female compared to 110 Male, and 4 listed as other. The sample is somewhat female-skewed in both groups, but especially in the patient group (54.7% gen and 59.6% diagnosed). The skewed distribution risks training a model that learns patterns that are more representative of females than males, simply due to a lack of representation of symptomatic males or healthy females.

#2: Measurement & labeling bias

Psychiatric symptom scales are useful, but they are not perfect measurements of mental health. Self-report questionnaires can be influenced by mood, interpretation of questions, willingness to disclose symptoms, or clinician rating differences, which may introduce noise into the symptom severity labels.

#3: Overfitting / Site-related bias

Our model uses extracted imaging features rather than raw MRI images, but these features can still be influenced by scanner differences, participant motion, and other site-specific effects. Our exploratory analysis revealed Site 2 contains nearly twice as many patients (70) as Site 1 (38). This imbalance increases the risk that the model will associate site-related characteristics rather than learning meaningful patterns.

Mitigation Strategies

Addressing/Correcting representation imbalances

- Examine the dataset for demographic imbalance and missing data
- Use stratified sampling when splitting data into training, validation, and testing sets to ensure the true gender ratio is even across both GenPop and Patient clinical cohorts
- If some groups remain underrepresented, implement class-weighting to ensure the model penalizes errors on the underrepresented populations equally.
- Evaluate final model performance separately across demographic groups to audit for fair and equitable predictions.

Preprocessing, normalization, and validation

- Use careful preprocessing, feature normalization, and strong validation procedures such as cross-validation and a held-out test set to reduce overfitting and improve generalizability
- Compare behavioral-only, imaging-only, and multimodal models to better understand which data sources drive performance and to avoid overstating the role of brain imaging if simpler behavioral features explain most of the predictive signal.

Citations

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